CivicFix AI

Sources, References & Data Provenance

Group 8 | Supporting document | 29 October 2026

1. Data sources used in the project

1. NYC Open Data — 311 Service Requests (2020 to Present)

Official link: https://data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2020-to-Present/erm2-nwe9

Project use / scope: 2025 records filtered to Street Condition, Sidewalk Condition, Sewer, and Root/Sewer/Sidewalk Condition. Source of complaint categories, dates, locations, and recurrence labels. Local file: nyc311_infra_2025.csv. This is a dynamically updated dataset; extraction counts may differ by date.

2. NOAA NCEI — Global Historical Climatology Network Daily (GHCN-Daily)

Official link: https://www.ncei.noaa.gov/products/land-based-station/global-historical-climatology-network-daily

Project use / scope: New York daily precipitation, Central Park station USW00094728, used to associate precipitation with NYC complaints. Local file: nyc_centralpark_2025.csv.

3. Open-Meteo — Historical Weather API

Official link: https://open-meteo.com/en/docs/historical-weather-api

Project use / scope: Historical precipitation for a selected Istanbul coordinate, 1–20 October 2026. Local outputs: istanbul_rain_oct2026.csv and istanbul_rain.png. Gridded model/reanalysis weather, not a citywide gauge observation.

4. Copernicus — Copernicus Digital Elevation Model GLO-30

Official link: https://dataspace.copernicus.eu/explore-data/data-collections/copernicus-contributing-missions/copernicus-digital-elevation-model

Project use / scope: Elevation raster downloaded through OpenTopography (COP30), used for pilot terrain/elevation and slope analysis. Local input archive: rasters_COP30.tar. Derived graphic: istanbul_terrain.png. A DEM-derived terrain indicator is not a validated flood probability.

5. OpenTopography — Global DEM access portal

Official link: https://portal.opentopography.org/

Project use / scope: Access and download interface used to obtain the Copernicus DEM subset for the Istanbul pilot region.

6. Istanbul Metropolitan Municipality — Open Data Portal

Official link: https://data.ibb.gov.tr/

Project use / scope: Local source of the uploaded Istanbul dam rainfall and daily water consumption spreadsheet. This historical dam dataset is contextual information; it is not evidence of the October 2026 flood event and is not equivalent to street-level rainfall.

2. Analytical outputs and their origins

Output	Derived from
NYC model ROC-AUC and top-decile precision	NYC 311 records + NOAA precipitation; computed by CivicFix EDA code, not statistics published by NYC Open Data.
NYC rainfall–complaint plot	NYC 311 + NOAA daily precipitation.
Istanbul October 2026 rainfall chart	Open-Meteo selected-location historical precipitation.
Istanbul elevation/slope pilot	Copernicus GLO-30 via OpenTopography; locally derived raster calculations.

3. Sources proposed for future work (not completed validation)

OpenStreetMap — https://www.openstreetmap.org/
Proposed purpose: Stream networks / drainage proxies; buried streams may be absent.

Copernicus Sentinel-1 — https://dataspace.copernicus.eu/
Proposed purpose: Potential flood-extent verification; no completed satellite-based validation is claimed.

Turkish complaint records / social media — Not yet established
Proposed purpose: Requires provenance, permission where relevant, anonymization and independent manual labels.

4. Reproducibility and limitations

- The NYC model is evaluated on NYC complaints. Its performance is not an Istanbul accuracy estimate.
- A single Istanbul weather grid point cannot represent precipitation across all districts.
- Elevation and slope alone do not establish surface-water flow or observed inundation; hydrological processing and ground-truth data would be needed.
- The IBB dam spreadsheet should not be presented as 2026 street-level flood evidence.
- Open datasets may be revised. Keep the original downloaded files, extraction dates, code, and data filters with the submission.
- Data access dates for individual historical downloads were not recorded consistently; this reference register was prepared on 29 October 2026.

5. Suggested citation wording

“NYC 311 service requests were obtained from NYC Open Data (dataset erm2-nwe9), daily New York precipitation from NOAA GHCN-Daily, Istanbul historical precipitation from Open-Meteo, and terrain elevations from Copernicus DEM GLO-30 via OpenTopography. All model metrics and derived maps are outputs of the authors’ analysis, not official source statistics.”